



# Division Decoded

The mission where the leftover needs a decision.

## THE BIG QUESTION

When you have a remainder, what do you do with it — and who decides?

## HANDS-ON PROJECT

### Snack Run

Split real quantities among real people.  
Defend every remainder decision out loud.

## FLUENCY GAME

### Remainder Race

Deal cards, divide, score the leftover. Biggest remainder takes the round.



AXIOM • MISSION COMMANDER, NINE TOURS

“Brock — the leftover is not a mistake. It's information. Most people never learn to read it.”

# Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

**This is not a test.** Nothing here counts for a grade. It exists to delete work you don't need. If you don't know something, write "not yet" and move on — that's a useful answer.

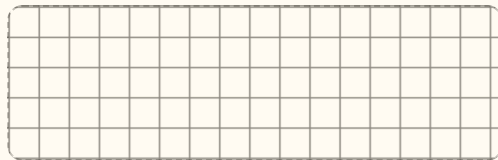
## STRAND A FACTS

A1  $42 \div 6 =$  \_\_\_\_\_

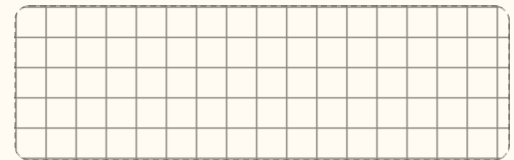
A2  $56 \div 8 =$  \_\_\_\_\_

## STRAND B MISSING SIDE

B1 Draw  $84 \div 4$  as a rectangle.



B2  $156 \div 6 =$  \_\_\_\_\_



## STRAND C REMAINDERS

C1  $94 \div 4 =$  \_\_\_\_\_ r \_\_\_\_\_

\_\_\_\_\_

C2 26 kids, vans hold 6. How many vans? Why?

\_\_\_\_\_

## STRAND D LONG DIVISION

D1  $372 \div 3 =$  \_\_\_\_\_

\_\_\_\_\_

D2  $845 \div 4 =$  \_\_\_\_\_ r \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## STRAND E DIVISIBILITY

E1 Which of 2, 3, 5, 9 divide 234?

\_\_\_\_\_

E2 Does 7 divide 91? How do you know?

\_\_\_\_\_

## STRAND F REASONING

F1 100 stickers, 7 kids. What happens to the extras?

\_\_\_\_\_

F2  $\square 6 \div 4 = 24$   $\square =$  \_\_\_\_\_

\_\_\_\_\_

# Skip Chart

Score the pre-assessment, then cross weeks off the calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

| STRAND                   | ITEMS    | COVERS                               | SCORE      | IF 2/2, REMOVE                                                                |
|--------------------------|----------|--------------------------------------|------------|-------------------------------------------------------------------------------|
| <b>A • Facts</b>         | A1<br>A2 | Division facts to 81                 | --- /<br>2 | Warm-Up tier drops to 3 items all unit                                        |
| <b>B • Missing side</b>  | B1<br>B2 | Week 1 • lessons 1.1–1.2             | --- /<br>2 | Days 1.1 and 1.2 — keep Friday enrichment                                     |
| <b>C • Remainders</b>    | C1<br>C2 | Week 1 • 1.3 and<br>Week 2 • 2.1–2.3 | --- /<br>2 | Days 2.1–2.3. Never skip 2.4 — remainder-as-answer is new to almost everyone. |
| <b>D • Long division</b> | D1<br>D2 | Week 1 • 1.4 and<br>Week 3           | --- /<br>2 | Week 3 days 3.1–3.3. Keep 3.4, the multiply-back check.                       |
| <b>E • Divisibility</b>  | E1<br>E2 | Week 4 • lessons 4.1–4.2             | --- /<br>2 | Days 4.1–4.2. Never skip 4.4 (two-digit divisors).                            |
| <b>F • Reasoning</b>     | F1<br>F2 | Week 2 • 2.1–2.4<br>and Week 5 • 5.1 | --- /<br>2 | Nothing. Raise the Challenge tier instead.                                    |

## Where the saved time goes

Never back into the calendar. Compacted time is spent on the Challenge tier, the Snack Run project, or an early start on Mission 03. Write the reallocation here so it doesn't quietly evaporate:

.....  
 .....

## Revised Mission 02 plan

WEEK 1 ☐ full ☐ compressed ☐ skip

WEEK 2 ☐ full ☐ compressed ☐ skip

WEEK 3 ☐ full ☐ compressed ☐ skip

WEEK 4 ☐ full ☐ compressed ☐ skip

WEEK 5 ☐ full ☐ compressed ☐ skip

# Two Kinds of Division

Deal them into rows, or deal them into groups. Same picture, two questions.

## Warm-Up 6 ITEMS • 5 MINUTES • FACTS YOU ALREADY OWN

$42 \div 6$   
\_\_\_\_\_

$56 \div 8$   
\_\_\_\_\_

$63 \div 9$   
\_\_\_\_\_

$36 \div 4$   
\_\_\_\_\_

$72 \div 8$   
\_\_\_\_\_

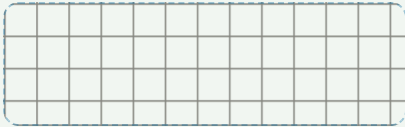
$45 \div 5$   
\_\_\_\_\_

$42 \div 6$  **Sharing:** 6 equal rows — how many in a row? **Grouping:** groups of 6 — how many groups?

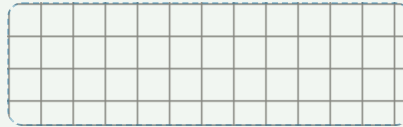
Both: **7**

## Core DRAW THE RECTANGLE. SPLIT THE AREA INTO FRIENDLY PIECES.

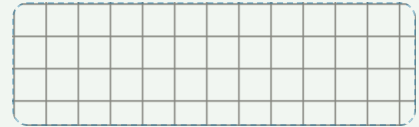
1.  $84 \div 4$  = \_\_\_\_\_



2.  $96 \div 6$  = \_\_\_\_\_



3.  $75 \div 5$  = \_\_\_\_\_



4.  $90 \div 6$  = \_\_\_\_\_



5.  $78 \div 3$  = \_\_\_\_\_



For number 5, write the two pieces you split 78 into:

\_\_\_\_\_

## ★ Challenge NOT GRADED

**C1.** A rectangle has area 96 and one side 6. How long is the other side? Show how you found it.

**C2.** 72 pencils go into boxes of 8. How many boxes? Is that a sharing question or a grouping question?

**C3.** Write two stories for  $48 \div 8$  — one sharing, one grouping. They should sound completely different.



**AXIOM:** Division is the same rectangle read from a different corner. Say the multiplication fact out loud first.

# The Missing Side

$156 \div 6$  is hard. Twenty sixes and six sixes are easy. That's the whole trick.

## Warm-Up

$24 \div 3$   
\_\_\_\_\_

$54 \div 6$   
\_\_\_\_\_

$81 \div 9$   
\_\_\_\_\_

$32 \div 8$   
\_\_\_\_\_

$49 \div 7$   
\_\_\_\_\_

$60 \div 5$   
\_\_\_\_\_

### WORKED EXAMPLE

$156 \div 6$

20 sixes  
120

6 sixes  
36

$20 + 6 = 26$

Add the *widths*, not the areas. The areas were 120 and 36 — those already got used up.

## Core TWO CHUNKS EACH. WRITE BOTH BEFORE YOU ADD THEM.

1.  $128 \div 4$  ..... = \_\_\_\_\_

2.  $156 \div 6$  ..... = \_\_\_\_\_

3.  $144 \div 3$  ..... = \_\_\_\_\_

4.  $175 \div 5$  ..... = \_\_\_\_\_

5.  $168 \div 7$  ..... = \_\_\_\_\_

## Challenge

**C1.** A rectangle has area 252 and one side 7. Find the other side using two chunks.

.....

**C2.** Which is bigger,  $144 \div 6$  or  $168 \div 8$ ? Decide first, then check.

.....

**C3.** Divide 96 by 6 using only chunks of ten sixes. How many steps did it take? Now do it in one step.

# When It Doesn't Fit

The quotient and the remainder are two different answers. Read which one is being asked for.

## Warm-Up REMAINDER ONLY — DON'T WRITE THE QUOTIENT

$17 \div 5$   
\_\_\_\_\_

$23 \div 4$   
\_\_\_\_\_

$38 \div 6$   
\_\_\_\_\_

$50 \div 7$   
\_\_\_\_\_

$29 \div 3$   
\_\_\_\_\_

$45 \div 8$   
\_\_\_\_\_

THE PIECE THAT WON'T FIT

$94 \div 4 = 23 \text{ remainder } 2$

23 columns of 4 = 92

r2

The dashed sliver is too short to make a full column. Draw it every time — don't just write "r2".

## Core BOTH ANSWERS EVERY TIME, THEN THE CHECK

1.  $17 \div 5$

q \_\_\_\_\_

r \_\_\_\_\_

check:  $5 \times \underline{\quad} + \underline{\quad} = 17$ 

2.  $94 \div 4$

q \_\_\_\_\_

r \_\_\_\_\_

check:  $4 \times \underline{\quad} + \underline{\quad} = 94$ 

3.  $137 \div 6$

q \_\_\_\_\_

r \_\_\_\_\_

check:  $6 \times \underline{\quad} + \underline{\quad} = 137$ 

4.  $83 \div 9$

q \_\_\_\_\_

r \_\_\_\_\_

check:  $9 \times \underline{\quad} + \underline{\quad} = 83$ 

## Challenge

**C1.** 26 kids, vans that hold 6. How many vans? What happened to the leftover?

.....

**C2.** 26 cookies, 6 friends, whole cookies only. How many each? What happened to the leftover this time?

.....

**C3.** Same numbers, different answers. Write the rule you'd give another kid for deciding which to do.

.....

# Chunks Get Tidy

Three-digit dividends. Take the biggest chunk you can see, then finish what's left.

 Warm-Up FRIENDLY HUNDREDS

$$120 \div 4$$

$$150 \div 5$$

$$240 \div 6$$

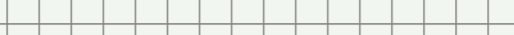
$$350 \div 7$$

$$180 \div 9$$

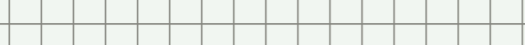
$$160 \div 8$$

 **Core** SHOW EVERY CHUNK YOU TOOK. ESTIMATE FIRST.

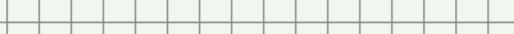
1.  $372 \div 3$  = \_\_\_\_\_



2.  $465 \div 5$  = \_\_\_\_\_



3.  $728 \div 8$  = \_\_\_\_\_



4.  $594 \div 6$  = \_\_\_\_\_

5.  $952 \div 7 = \underline{\hspace{2cm}}$

Estimate before you start — roughly how big will the answer be?

### ★ Challenge

**C1.**  $845 \div 4 = \underline{\hspace{2cm}}$  r  $\underline{\hspace{2cm}}$ . Show the chunks you used.

**C2.** Check your answer to  $952 \div 7$  by multiplying it back. Write the check as one equation.

**C3.** Invent a division with a three-digit dividend that comes out exactly, with no remainder. Prove it does.

# Remainder Race & Puzzles

No Warm-Up today. Play the game first, then take the puzzles as slowly as you like.

## Remainder Race • two players, one deck

Strip the face cards. Each player turns two cards to make a two-digit number, then rolls a die for the divisor.

Divide. **Your score for the round is the remainder.** Ten rounds, highest total wins. A clean division scores zero — which means the safe-looking answer is the worst one.

After three rounds, stop and answer this: which divisor gives you the best chance of a big score, and why?

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## Puzzles REASON IT OUT — NO GUESSING

P1  $100 \div 7 = \underline{\quad\quad} \text{ r } \underline{\quad\quad}$

P2  $7 \times \square = 91$   $\square = \underline{\quad\quad}$

P3  $\square \div 6 = 14$   $\square = \underline{\quad\quad}$

P4  $9 \times \square = 108$   $\square = \underline{\quad\quad}$

P5 Smallest number over 50 that 7 divides exactly: \_\_\_\_\_

P6  $\square 6 \div 4 = 24$   $\square = \underline{\quad\quad}$

## Two decisions, one division

Both of these are  $53 \div 8$ . The answers are different. Write each one and say why.

**a.** 53 people at a party, tables seat 8. How many tables?

**b.** You have \$53 and tickets cost \$8. How many tickets?

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.....

## The real question

Divide lots of different numbers by 5 and list every remainder you get. Which remainders are possible? Which are impossible? Explain why a remainder of 5 can never happen.

.....

.....



# The Rest of the Mission

Same three-tier shape as Week 1, every day. Objectives and problem sets outlined here; full worksheets follow the Week 1 pattern.

## Week 2 • Remainders

THE BIG QUESTION GETS ANSWERED THIS WEEK

### 2.1 Mon

Round up. Vans, boats, buses — when two leftovers force one more.

### 2.2 Tue

Drop it. Whole cookies each; the leftover stays on the plate.

### 2.3 Wed

Share it out. The remainder becomes a fraction of the thing.

### 2.4 Thu

The remainder is the answer: clocks, patterns, every-nth questions.

### Fri

Snack Run planning. Real quantities, every decision written down.

## Week 3 • Long Division

MID-UNIT QUIZ FRIDAY • 85% TO ADVANCE

### 3.1 Mon

Under the bar. Partial quotients shrink into the standard layout.

### 3.2 Tue

Three-digit dividends. Estimate before you start, every time.

### 3.3 Wed

Zeros in the quotient.  $618 \div 6$  catches everybody once.

### 3.4 Thu

Check by multiplying back. Self-marking from here on.

### Fri

Mid-unit quiz, 8 items, Weeks 1–3.

## Week 4 • Divisibility & Big Numbers

NEVER SKIP THE TWO-DIGIT DIVISORS

### 4.1 Mon

Tests for 2, 5, 10. Why the last digit is enough on its own.

### 4.2 Tue

Tests for 3 and 9 — and why adding the digits works at all.

### 4.3 Wed

Four-digit dividends.  $3,472 \div 8$  without flinching.

### 4.4 Thu

Two-digit divisors. Estimate, adjust, estimate again.

### Fri

Perfect numbers:  $6 = 1 + 2 + 3$ . Hunt the next one.

## Week 5 • Proof & Project

UNIT TEST FRIDAY • BADGE AWARDED

### 5.1 Mon

Missing-digit division. Reason backwards from the quotient.

### 5.2 Tue

Snack Run: split real quantities among real people.

### 5.3 Wed

Remainder defence. Justify every leftover call out loud.

### Thu

Error journal sweep. Fix only what repeats.

### Fri

Mission 02 test: 12 items + one explanation.

### MATERIALS, WHOLE UNIT

Base-ten blocks • counters or dried beans • two dice • deck of cards • graph paper ( $\frac{1}{4}$  in) • the Snack Run shopping list • the error journal

### WEEKLY CHECK

Five items every Friday, mixed tiers. Below  $\frac{4}{5}$  means Monday is a reteach, not new material.

### THE MOST COMMON STALL

A right quotient with the wrong decision about the remainder. Ask “what were you dividing?” before you mark anything wrong.

# Mid-Unit Quiz

8 items · Weeks 1–3 · 7 of 8 keeps you flying

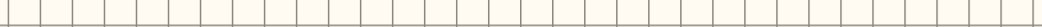
\_\_\_ / 8

1.  $63 \div 9 =$

**2.** Draw  $84 \div 4$  as a rectangle. Label both chunks and write the answer.



**3.**  $156 \div 6$ . Any method you like — but show the method.



4.  $94 \div 4 = \underline{\hspace{2cm}}$  r  $\underline{\hspace{2cm}}$ . Then check it.

.....

**5.** 29 tennis balls, tubes hold 4. How many tubes to store them all?

=====

=====

6.  $372 \div 3 = \underline{\hspace{2cm}}$

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7. Which of 2, 3, 5, 9 divide 234? Name the test you used.

.....

.....

## CHALLENGE ITEM

8.  $\square 6 \div 4 = 24$   $\square =$  \_\_\_\_\_

Clue you used first:

[illegible]

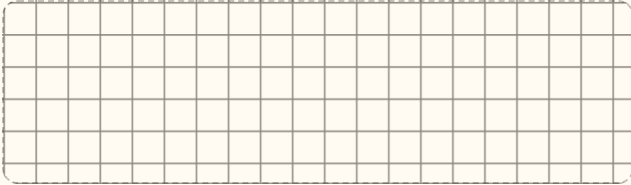
# Mission 02 Test

12 items + one explanation · part 1 of 2

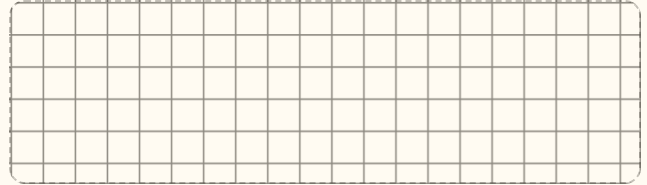
1.  $72 \div 8 =$  .....

2.  $54 \div 6 =$  .....

3.  $96 \div 4$  with a rectangle.  $=$  \_\_\_\_\_



4.  $175 \div 5$ , show your chunks.  $=$  \_\_\_\_\_



5.  $137 \div 6 =$  \_\_\_\_\_ r \_\_\_\_\_



6.  $465 \div 5$ , any method.  $=$  \_\_\_\_\_



7.  $3,472 \div 8$ . Estimate first, then compute.

.....

8. 47 kids on a field trip, buses hold 9. How many buses?

.....

.....

9. You have \$47. Comics cost \$9. How many can you buy?

.....

.....

# Reasoning & the Big Question

10. Which of 2, 3, 5, 9, 10 divide 450? Show the test you used for each.

.....

.....

11. Someone says  $728 \div 8 = 91$ . Are they right? Prove it without dividing again.

.....

.....

CHALLENGE ITEM

12.  $1\boxed{5} \div 5 = 23$   $\boxed{\phantom{0}} = \underline{\hspace{2cm}}$

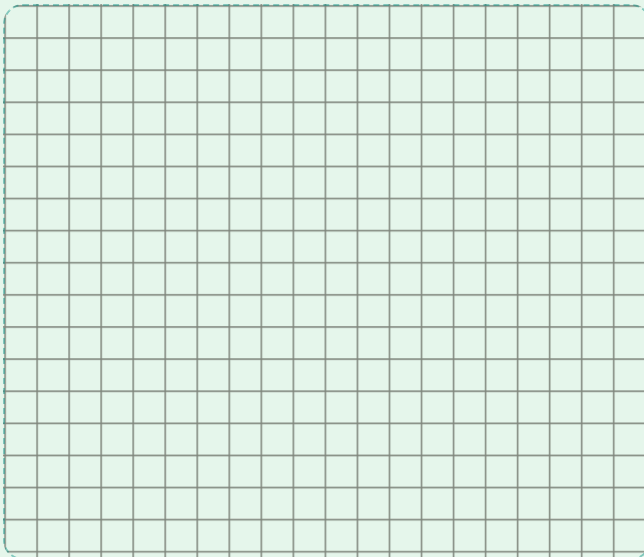
First clue:

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EXPLAIN YOUR THINKING • WORTH 4 POINTS

When you have a remainder, what do you do with it – and who decides?

Take  $26 \div 6$ . Write two different real situations that give two different right answers, and say what made them different. Draw the leftover in at least one. Spelling doesn't count. Reasoning does.



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TROPHY BAND • CIRCLE ONE

## 85–100%

### Mission Commander

Leftover Badge earned. Mission 03 opens — and Challenge becomes your Core.

## 70–84%

### Flight Engineer

Two targeted days on the flagged items, then retake those items only.

## Below 70%

### Ground Crew

Rebuild Week 1 with counters before returning to the long-division bar.

# Key A • Pre-Assessment, Lessons 1.1–1.2

## Pre-Assessment

A1 7    A2 7

B1 rectangle 4 by 21, split  $80 + 4 \rightarrow 21$

B2  $120 + 36 \rightarrow 20 + 6 = 26$

C1 23 r2

C2 5 vans — 4 vans hold 24, two kids still need seats

D1 124

D2 211 r1

E1 2, 3, 9 (not 5)

E2 Yes,  $7 \times 13 = 91$

F1 14 each, 2 left — accept any sensible fate for the 2

F2  $\square = 9$  ( $96 \div 4 = 24$ )

**Scoring note.** C2 and F1 are about the decision, not the arithmetic. A correct quotient with no reasoning scores half. Mark the strand partial and keep Week 2.

## Lesson 1.1 • Two Kinds of Division

Warm-Up  $7 \cdot 7 \cdot 7 \cdot 9 \cdot 9 \cdot 9$

Core

1.  $84 \div 4 = 20 + 4 = 21$

2.  $96 \div 6 = 10 + 6 = 16$

3.  $75 \div 5 = 10 + 5 = 15$

4.  $90 \div 6 = 10 + 5 = 15$

5.  $78 \div 3 = 20 + 6 = 26$  (60 and 18)

**Lesson 1.1 Challenge.** C1 • 16 ( $96 \div 6$ ; accept  $60 \div 6 + 36 \div 6$ ). C2 • 9 boxes; this is grouping — the group size is known and the number of groups is not. C3 • accept any two stories where one shares among a known number of people and the other makes groups of a known size, e.g. “48 stickers shared by 8 kids” versus “48 stickers into bags of 8.”

## Lesson 1.2 • The Missing Side

Warm-Up  $8 \cdot 9 \cdot 9 \cdot 4 \cdot 7 \cdot 12$

Core (chunks • answer)

1.  $120 \div 4 = 30$ ,  $8 \div 4 = 2 \rightarrow 32$

2.  $120 \div 6 = 20$ ,  $36 \div 6 = 6 \rightarrow 26$

3.  $120 \div 3 = 40$ ,  $24 \div 3 = 8 \rightarrow 48$

4.  $150 \div 5 = 30$ ,  $25 \div 5 = 5 \rightarrow 35$

5.  $140 \div 7 = 20$ ,  $28 \div 7 = 4 \rightarrow 24$

**Challenge.** C1 • 36 ( $210 \div 7 = 30$ , then  $42 \div 7 = 6$ ). C2 •  $144 \div 6 = 24$  is bigger;  $168 \div 8 = 21$ . Strong reasoning notices that 168 is only a little more than 144 while 8 is much more than 6. C3 • ten sixes four times gets to 240 — too far; the honest answer is one chunk of ten (60) then six more sixes, or in one step sixteen sixes. Accept any correct chunking; the point is that bigger chunks mean fewer steps.

**Watch for:** adding the areas instead of the widths — writing  $156 \div 6 = 156$ . If that happens, go back to counters for one day.

# Key B • Lessons 1.3–1.4, Enrichment, Quiz

## Lesson 1.3 • When It Doesn't Fit

**Warm-Up** (remainders) 2 • 3 • 2 • 1 • 2 • 5

**Core** (quotient • remainder • check)

1. 3 r2 •  $5 \times 3 + 2 = 17$
2. 23 r2 •  $4 \times 23 + 2 = 94$
3. 22 r5 •  $6 \times 22 + 5 = 137$
4. 9 r2 •  $9 \times 9 + 2 = 83$

**Challenge.** C1 • 5 vans; the leftover forced an extra van. C2 • 4 cookies each; the leftover was dropped. C3 • accept any rule that names the situation as the decider — “if the leftover still needs looking after, round up; if it just sits there, drop it.”

## Lesson 1.4 • Chunks Get Tidy

**Warm-Up** 30 • 30 • 40 • 50 • 20 • 20

1.  $300 \div 3 + 60 \div 3 + 12 \div 3 = 124$
2.  $450 \div 5 + 15 \div 5 = 93$
3.  $720 \div 8 + 8 \div 8 = 91$
4.  $540 \div 6 + 54 \div 6 = 99$
5.  $700 \div 7 + 210 \div 7 + 42 \div 7 = 136$

**Challenge.** C1 • 211 r1 ( $800 \div 4 = 200$ ,  $44 \div 4 = 11$ , 1 left). C2 •  $7 \times 136 = 952$ , so the check closes exactly. C3 • any dividend built as divisor  $\times$  whole number; the proof is the multiplication written back.

## Friday • Remainder Race

**Puzzles** P1 14 r2 • P2  $\square = 13$  • P3  $\square = 84$

P4  $\square = 12$  • P5 56 • P6  $\square = 9$

**Game question.** A divisor of 6 gives the best shot at a big score — the possible remainders run 0 to 5, so the ceiling is highest. Dividing by 2 can only ever score 0 or 1. Accept any answer that spots that the biggest possible remainder is always one less than the divisor.

**Two decisions.** a • 7 tables (six seat 48, five people still standing). b • 6 tickets, with \$5 left over. Same division, opposite treatment of the leftover.

**The real question.** Dividing by 5, the only possible remainders are 0, 1, 2, 3, 4. A remainder of 5 is impossible because another whole 5 would fit — the quotient was simply too small. Accept any wording that reaches “the remainder is always smaller than the divisor.”

## Mid-Unit Quiz • out of 8

1. 7
2.  $84 \div 4 = 20 + 1 = 21$
3.  $156 \div 6 = 20 + 6 = 26$
4. 23 r2 • check  $4 \times 23 + 2 = 94$
5. 8 tubes (7 hold 28, one ball over)
6. 124
7. 2, 3, 9 — even; digits sum to 9
8.  $\square = 9$  ( $96 \div 4 = 24$ )

**7/8 or better → continue.** Below that, reteach only the items missed. A miss on 4 or 5 means remainders, not division — spend the two days on Week 2, not Week 1.

1. 9
2. 9
3.  $96 \div 4 = 20 + 4 = 24$
4.  $175 \div 5 = 30 + 5 = 35$
5. 22 r5
6.  $465 \div 5 = 90 + 3 = 93$
7. estimate  $\approx 400 \cdot 434$
8. 6 buses (five hold 45, two kids over)
9. 5 comics, \$2 left
10. all five - 450 is even, digits sum to 9, ends in 0
11. Yes -  $8 \times 91 = 728$ , remainder 0
12.  $\square = 1$  ( $115 \div 5 = 23$ )

**Total 16 points.** 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. Item 12 missed alone never blocks the badge. Items 8 and 9 missed together means Week 2 gets revisited before Mission 03.